Thermodynamic description of the Al–Si system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Al–Si binary system as calculated with pycalphad from the SGTE dataset AlSi-12Tan.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Al) + (Si), lies at 850 K.

1. Introduction

The Al–Si system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as simple eutectic. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of Y. Tang, Y. Du, L. Zhang, X. Yuan, G. Kaptay, Thermochim. Acta, 527, 131-42(2012). The calculations use pycalphad 0.11.2 at 1 atm.

Pure Al melts at 935 K. Pure Si melts at 1687 K.

2. Phase diagram

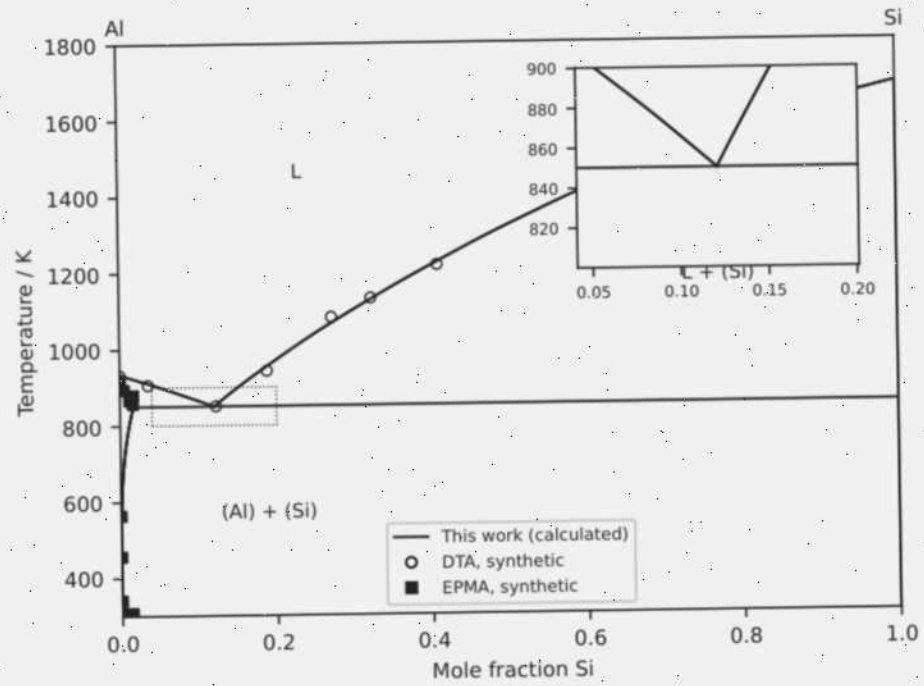


Fig. 1. Calculated Al–Si phase diagram with synthetic experimental-style points (inset: enlarged view near the lowest invariant).

At 850 K the eutectic L → (Al) + (Si) takes place; the L phase then contains x(Si) = 0.121.

Table 1. Invariant reactions in the Al–Si system (calculated).

Reaction	Type	T / K	Compositions (x(Si))		
			phase 1	phase 2	phase 3
L → (Al) + (Si)	eutectic	850.2	0.121	0.017	1.000

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Thermodynamic properties

At 1750 K the integral mixing enthalpy of the liquid has its minimum, −3462 J/mol, at x(Si) = 0.520.

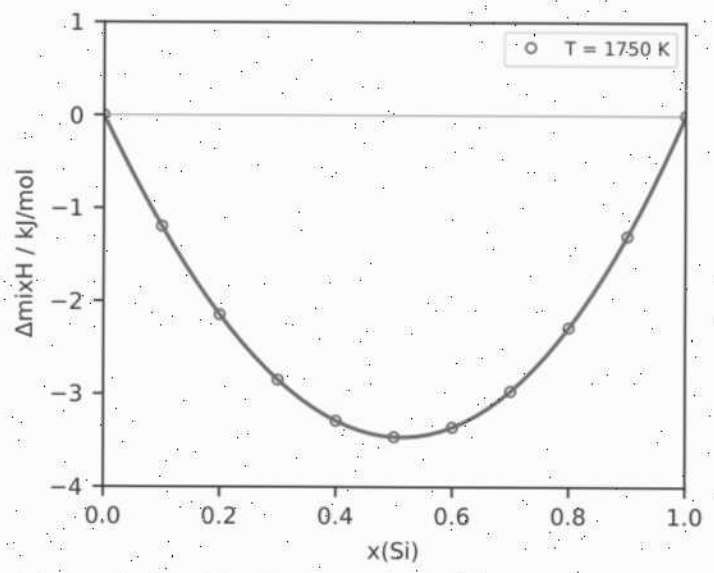


Fig. 2. Calculated mixing enthalpy of the liquid at 1750 K.

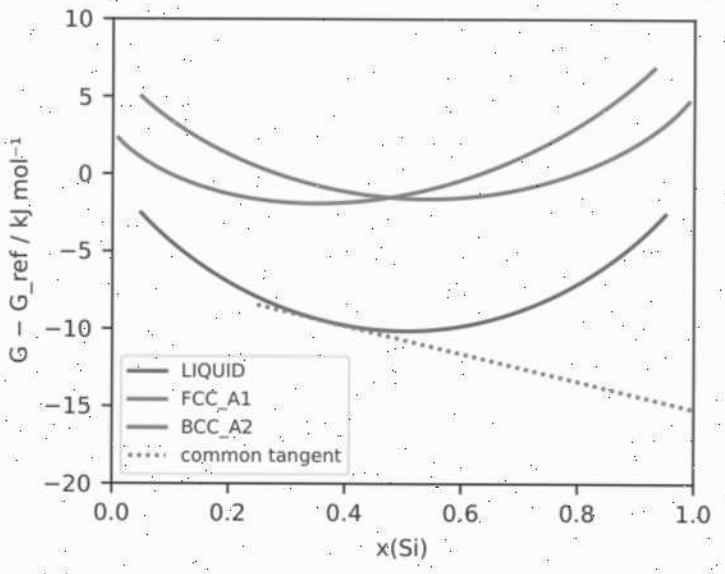


Fig. 3. Calculated Gibbs energy curves at 1180 K.

Table 2. Redlich–Kister parameters of the liquid (Al–Si).

	Liquid: L(ν) = a + b·T + c·T·ln T		
ν	**a**	**b**	**c**
L0	-13842.105	0.45733	0.0
L1	717.10348	-0.2012	0.0

Values as printed in the TDB file.

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.